Energy-Efficient Control of an Inverted Pendulum:

Nonlinear Dynamic Modeling via Lagrange's Method

Project Report
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1 System Overview & Generalized Coordinates

We consider an inverted pendulum mounted on a cart that moves along a horizontal track. The system consists of a cart of mass M, a pendulum of mass m with its center of mass located at a distance l from the pivot, and an applied horizontal control force u acting directly on the cart. Gravity acts downwards with acceleration g.

The system is defined using two generalized coordinates:

$$q = \begin{bmatrix} x \\ \theta \end{bmatrix} \tag{1}$$

where:

- x: horizontal position of the cart.
- θ: pendulum angular displacement relative to the upright vertical position ($\theta = 0$ corresponds to the upright equilibrium).

2 Kinematic Equations

The position and velocity vectors for both the cart and the pendulum center of mass are derived as follows:

2.1 Cart Kinematics

$$x_c = x, \quad y_c = 0 \tag{2}$$
$$\dot{x}_c = \dot{x}, \quad \dot{y}_c = 0 \tag{3}$$

2.2 Pendulum Center of Mass Kinematics

$$x_p = x + l\sin\theta \tag{4}$$
$$y_p = l\cos\theta \tag{5}$$

Taking the time derivatives of the position coordinates:

$$\dot{x}_p = \dot{x} + l\dot{\theta}\cos\theta \tag{6}$$
$$\dot{y}_p = -l\dot{\theta}\sin\theta \tag{7}$$

3 Kinetic and Potential Energy Calculations

3.1 Kinetic Energy

The total kinetic energy T of the system is the sum of the kinetic energy of the cart T_c and the pendulum T_p:

$$T = T_c + T_p \tag{8}$$

1. **Cart Kinetic Energy (T_c):**

$$T_c = \frac{1}{2} M \dot{x}^2 \tag{9}$$

2. **Pendulum Kinetic Energy (T_p):** The velocity magnitude squared of the pendulum mass center (v_p^2) is given by:

$$\begin{aligned} v_p^2 &= \dot{x}_p^2 + \dot{y}_p^2 && (10)\\ &= \left(\dot{x} + l\dot{\theta}\cos\theta\right)^2 + \left(-l\dot{\theta}\sin\theta\right)^2 && (11)\\ &= \dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta && (12)\\ &= \dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2\left(\cos^2\theta + \sin^2\theta\right) && (13)\\ &= \dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2 && (14) \end{aligned}$$

Therefore, the kinetic energy of the pendulum is:

$$T_p = \frac{1}{2} m \left(\dot{x}^2 + 2l\dot{x}\dot{\theta}\cos\theta + l^2\dot{\theta}^2\right) \tag{15}$$

3. **Total Kinetic Energy (T):**

$$T = \frac{1}{2}(M+m)\dot{x}^2 + ml\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}ml^2\dot{\theta}^2 \tag{16}$$

3.2 Potential Energy

Taking the track position $y = 0$ as the zero-potential reference:

$$V = mgy_p = mgl\cos\theta \tag{17}$$

4 Lagrangian Formulation

The Lagrangian $L = T - V$ of the system is defined as:

$$L = \frac{1}{2}(M+m)\dot{x}^2 + ml\dot{x}\dot{\theta}\cos\theta + \frac{1}{2}ml^2\dot{\theta}^2 - mgl\cos\theta \tag{18}$$

5 Euler-Lagrange Equations of Motion

The general form of the Euler-Lagrange equation for coordinate q_i is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i \tag{19}$$

where $Q_x = u$ (external force on cart) and $Q_\theta = 0$ (no direct torque on pivot).

5.1 Equation for x ($q_1 = x$)

$$\frac{\partial L}{\partial \dot{x}} = (M+m)\dot{x} + ml\dot{\theta}\cos\theta \tag{20}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = (M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta \tag{21}$$

$$\frac{\partial L}{\partial x} = 0 \tag{22}$$

Yielding the first equation of motion:

$$(M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = u \tag{23}$$

5.2 Equation for θ ($q_2 = \theta$)

$$\frac{\partial L}{\partial \dot{\theta}} = ml\dot{x}\cos\theta + ml^2\dot{\theta} \tag{24}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = ml\ddot{x}\cos\theta - ml\dot{x}\dot{\theta}\sin\theta + ml^2\ddot{\theta} \tag{25}$$

$$\frac{\partial L}{\partial \theta} = -ml\dot{x}\dot{\theta}\sin\theta + mgl\sin\theta \tag{26}$$

Substituting into the Lagrange equation:

$$\Big(ml\ddot{x}\cos\theta - ml\dot{x}\dot{\theta}\sin\theta + ml^2\ddot{\theta}\Big) - \Big(-ml\dot{x}\dot{\theta}\sin\theta + mgl\sin\theta\Big) = 0 \tag{27}$$

Simplifying gives:

$$ml\ddot{x}\cos\theta + ml^2\ddot{\theta} - mgl\sin\theta = 0 \tag{28}$$

Dividing by ml:

$$l\ddot{\theta} + \ddot{x}\cos\theta - g\sin\theta = 0 \tag{29}$$

6 Nonlinear Equations of Motion

6.1 Coupled System

Combining equations (23) and (29):

$$\begin{cases} (M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = u \\ l\ddot{\theta} + \ddot{x}\cos\theta - g\sin\theta = 0 \end{cases} \tag{30}$$

6.2 Explicit Form for Numerical Integration

For state-space modeling and simulator integration (e.g., using `scipy.integrate.solve_ivp`), solving for $\ddot{x}$ and $\ddot{\theta}$ explicitly yields:

$$\ddot{x} = \frac{u + ml\dot{\theta}^2\sin\theta - mg\sin\theta\cos\theta}{M + m\sin^2\theta} \tag{31}$$

$$\ddot{\theta} = \frac{(M+m)g\sin\theta - u\cos\theta - ml\dot{\theta}^2\sin\theta\cos\theta}{l(M + m\sin^2\theta)} \tag{32}$$